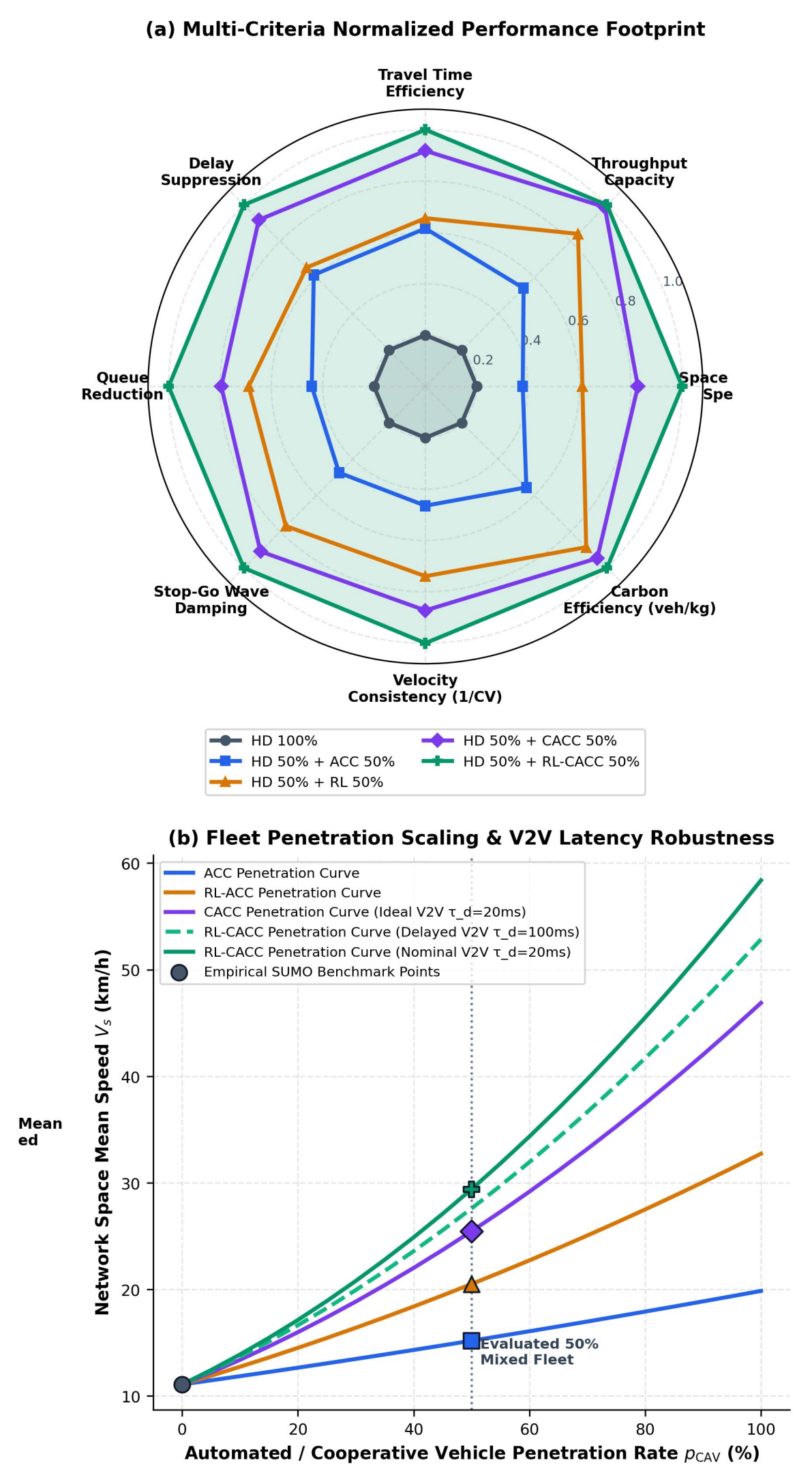

(a) Multi-Criteria Normalized Performance Footprint
Travel Time Efficiency
Throughput Capacity
Space Spe
Carbon Efficiency (veh/kg)
Velocity Consistency (1/CV)
Stop-Go Wave Damping
Queue Reduction
Delay Suppression
0.2
0.4
0.6
0.8
1.0
HD 100%
HD 50% + ACC 50%
HD 50% + RL 50%
HD 50% + CACC 50%
HD 50% + RL-CACC 50%
(b) Fleet Penetration Scaling & V2V Latency Robustness
ACC Penetration Curve
RL-ACC Penetration Curve
CACC Penetration Curve (Ideal V2V τ_d=20ms)
RL-CACC Penetration Curve (Delayed V2V τ_d=100ms)
RL-CACC Penetration Curve (Nominal V2V τ_d=20ms)
Empirical SUMO Benchmark Points
Evaluated 50% Mixed Fleet
Network Space Mean Speed $V_s$ (km/h)
Automated / Cooperative Vehicle Penetration Rate $p_{CAV}$ (%)
10
20
30
40
50
60
0
20
40
60
80
100